

open-provenance: An Open, Offline, User-Governed System for Verifiable Media Provenance and Durable Recovery

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Abstract

The marginal cost of producing convincing synthetic images, audio, and video has fallen to near zero, eroding society’s shared ability to answer a basic question: *did this actually happen?* We argue that the tractable response is not universal *detection* of synthetic media—which the literature shows to be brittle and adversarially fragile—but verifiable *provenance*: cryptographically signed, content-bound statements about how a piece of media came to be. We present **open-provenance**, an open-source system built on the C2PA standard that makes provenance verification (i) *offline*, requiring no network and no trusted third party at verification time; (ii) *user-governed*, so that the decision of *whom to trust* rests with the verifier rather than a central gatekeeper; and (iii) *honest*, with a conservative verdict model that never conflates an intact provenance chain with the truth of the depicted events, and never treats the absence of provenance as evidence of fabrication. We further study the central weakness of metadata-based provenance—that it is stripped by screenshots and social-media re-encoding—and prototype a *durable* recovery layer combining a perceptual-fingerprint registry with robust watermarking. On the Kodak natural-image suite, fingerprint-based recovery is robust to recompression and rescaling (ROC AUC = 1.0; mean Hamming distance 0.25 vs. 22.67 of 64 between matching and non-matching content under `resize-to-70% + JPEG-40 transport`). A *global* perceptual hash collapses under cropping (recovery near zero once half the area is removed); we resolve this with a *local-feature* fingerprint (ORB keypoints with RANSAC geometric verification) that sustains full recovery down to 50% retained area and 0.75 at 30% retained, where the perceptual hash recovers essentially nothing. The lightweight frequency-domain watermark fails by JPEG quality ≈ 50 . These results delimit, rather than oversell, what the open layer achieves. We are explicit about scope: the open ecosystem cannot read proprietary watermarks (e.g. SynthID), so durable recovery protects content that passes through participating signers, not arbitrary third-party media. Finally, we frame how such coverage scales through an open participation model—standard-C2PA interoperability (demonstrated by verifying third-party-signed content), a minimal signer on-ramp, and a single front door that returns an honest UNKNOWN for unsigned media. All code, the evaluation harness, and this report are released openly.

1 Introduction

Generative models now synthesize photorealistic media indistinguishable from captured reality [15]. This is not a niche concern: the ability to attribute media to a source underpins journalism, legal evidence, scientific record, elections, and ordinary interpersonal trust. Two broad technical responses exist. *Detection* asks a classifier whether a given artifact is synthetic; it is a moving target, content- and model-specific, and degrades under distribution shift and adversarial post-processing [10, 18].

Provenance instead attaches verifiable, signed statements to media at creation and edit time, shifting the question from “does this look fake?” to “what is this artifact’s documented history, and do I trust the party that attests to it?”

We take the provenance position, and observe that a provenance system intended as shared infrastructure must satisfy three properties that existing deployments only partially meet. First, **verification must be a public good**: it should require no account, no gatekeeper, and—critically—no network, so that a journalist in the field or a court clerk can verify locally. Second, **trust must be user-governed**: a verifier that ships a fixed list of “legitimate” signers is itself a central arbiter of authenticity, which is precisely the concentration of power a neutral system should avoid. Third, the system must be **epistemically honest**: it must distinguish “the cryptographic chain is intact” from “the depicted events are true,” and must never present the mere absence of provenance as evidence of manipulation.

Contributions.

1. An open verifier with a conservative, formally specified verdict and trust model that encodes the honesty properties above (§4), realized as both an offline CLI and a zero-backend in-browser WebAssembly application.
2. A user-governed trust mechanism in which the verifier cryptographically validates the signer’s certificate chain against anchors the *user* supplies, with no built-in default trust list (§4).
3. A study and open prototype of *durable* provenance recovery—surviving metadata stripping—combining a global perceptual-fingerprint registry, a crop-robust local-feature path, and robust watermarking, with a reproducible natural-image evaluation (§6).
4. An explicit threat model and an honest account of what open provenance can and cannot do, including the proprietary-watermark gap (§3, §7).
5. A participation and adoption model—standard-C2PA interoperability, a minimal signer on-ramp, and a single “verify any image” pipeline that returns an honest UNKNOWN for unsigned content rather than guessing (§8).

2 Background and Related Work

Provenance standards. The C2PA specification [3] defines a *manifest*: a set of assertions about an asset (capture device, edits, ingredients) bound to the asset’s pixels by cryptographic hashes, and signed using a COSE structure [17] whose signer is identified by an X.509 certificate chain [6], optionally timestamped via RFC 3161 [1]. The Content Authenticity Initiative [4] promotes adoption (“Content Credentials”). The IPTC `digitalSourceType` vocabulary [13] provides the standard marker for algorithmically generated media (`trainedAlgorithmicMedia`).

Watermarking. Classical invisible watermarking embeds a payload in transform coefficients (DWT/DCT/SVD) [7, 14]; such methods survive mild recompression and scaling but are fragile under strong compression and geometric distortion. Learned watermarks embed and detect payloads with neural networks and are markedly more robust: Stable Signature roots a watermark in a diffusion model’s decoder [11]; TrustMark targets arbitrary-resolution robustness for C2PA soft binding [2]; WAM localizes messages and resists cropping [16]; SynthID is a widely deployed but proprietary scheme [8, 12].

Fingerprinting. Perceptual hashes summarize low-frequency image structure into compact codes stable under recompression and scaling [19], enabling content matching against a registry. The combination of robust watermark, fingerprint, and manifest registry is the basis of “durable” Content Credentials [5].

3 Threat Model

Goals. For a verifier V presented with an asset a , the system should provide (G1) *integrity*—detect any modification of a after signing; (G2) *authenticity of the signer*—establish, relative to V ’s chosen trust anchors, which party signed a ; (G3) *recoverability*—when a ’s manifest has been stripped, recover it for content previously registered by a participating signer.

Adversary. We consider an adversary who may: strip or alter manifest metadata; recompress and rescale (lossy transport via screenshots and social platforms); crop or geometrically distort; substitute or self-sign with an attacker-controlled certificate (forge signer identity); and replay a valid manifest onto different content. The adversary does *not* possess the private keys of signers V trusts, and cannot break the underlying signature or hash primitives.

Non-goals. The system does *not* establish that the events depicted in a are true (a correctly signed photograph may still be staged), does *not* detect unmarked synthetic media, and does *not* read third-party proprietary watermarks. The absence of provenance is treated as *uninformative*, never as evidence of fabrication.

4 System Design

4.1 Honest verdict model

Given the validation outcome of a C2PA read, the verifier emits exactly one verdict $\in \{\text{VERIFIED}, \text{NO_CREDENTIALS},$

- **VERIFIED:** a manifest is present and all signature and content-hash checks pass. This asserts chain integrity only.
- **NO_CREDENTIALS:** no manifest is present. Formally uninformative about authenticity.
- **INVALID:** a manifest is present but a hash or signature check fails; the asset was altered after signing or the chain is broken.

Orthogonally, the verifier reports a trust status $\in \{\text{TRUSTED}, \text{UNTRUSTED}, \text{UNCHECKED}\}$. Let C be the signer’s certificate chain and $\mathcal{A}$ the user’s set of trust anchors. Then

$$\text{trust}(C, \mathcal{A}) = \begin{cases} \text{TRUSTED} & \text{if } C \text{ chains to some anchor in } \mathcal{A}, \\ \text{UNTRUSTED} & \text{if } \mathcal{A} \neq \emptyset \text{ and } C \text{ does not,} \\ \text{UNCHECKED} & \text{if } \mathcal{A} = \emptyset \text{ (or no trust engine).} \end{cases}$$

Crucially, **VERIFIED** is independent of trust: a cryptographically valid signature from an untrusted (e.g. self-signed) party yields $\langle \text{VERIFIED}, \text{UNTRUSTED} \rangle$, which the interface presents as a warning, never as an endorsement.

4.2 Offline verification and user-governed trust

Because the signer’s certificate travels inside the manifest, verification is *permissionless* and *local*: no network is contacted. The in-browser implementation disables remote-manifest fetching and OCSP, guaranteeing that no request leaves the device. Trust is supplied by the user as a set of PEM anchors; the toolkit validates C against $\mathcal{A}$ on-device. The system ships *no* default trust list, by design: adopting a published list (e.g. the C2PA list) is the user’s explicit choice, not an imposed policy.

4.3 Durable recovery layer

Metadata-based provenance is fragile: most platforms strip C2PA manifests. We therefore register, at signing time, a mapping from robust descriptors of the content to its manifest, using three complementary mechanisms. (i) A *global perceptual fingerprint* $\phi(a) \in \{0,1\}^{64}$ (a DCT perceptual hash) supports registry lookup by nearest Hamming distance; recovery declares a match iff $\min_e \text{Ham}(\phi(a'), \phi_e) \leq \tau$ for a threshold τ , framing recovery as a detection problem with an ROC over τ . This is robust to recompression and scaling but, being global, is not crop-invariant. (ii) A *local-feature fingerprint* (ORB keypoints matched with a Lowe ratio test and a RANSAC homography consensus) recovers cropped content: a crop retains many original keypoints, and geometric consensus confirms the source even when most of the frame is removed. (iii) An optional *robust watermark* embeds a short identifier directly in pixels, a self-contained pointer needing no registry. The recovered manifest is then re-verified by the normal offline path.

5 Implementation

The verifier is implemented twice over a single shared, dependency-free verdict classifier: a Node.js CLI using `c2pa-node`, and a zero-backend browser application that compiles the C2PA toolkit to WebAssembly so verification runs client-side and offline. A unified “verify any image” front door composes the layers (C2PA verification, then durable recovery, then an explicit UNKNOWN), and a participant SDK plus a reference upload handler let any entity sign-and-register or preserve-and-register received credentials. A signing tool emits manifests (optionally tagging generative content via `digitalSourceType`). The durable layer is implemented in Python over a DWT-DCT-SVD watermark [14], a DCT perceptual hash [19], and ORB keypoint matching with RANSAC homography verification for crop-robust recovery, with a registry supporting `register` and `recover`; the browser re-implements the *identical* perceptual hash (verified to agree bit-for-bit), so a registry built by the Python tools is usable in the browser for offline recovery. The evaluation harness (§6) regenerates every quantitative claim in this report from a single command.

6 Evaluation

Methodology. We evaluate on the Kodak natural-image suite [9] (24 true-colour 512×768 photographs with rich texture). A controlled synthetic corpus (sums of low-frequency sinusoids) served as a sanity baseline during development and gave conclusions consistent with those below for recompression and scaling; because smooth synthetic content is favourable to perceptual hashing, we report only the natural-image results here. Realistic transport includes JPEG recompression, rescaling, and—critically—*center cropping*, the geometric distortion that frequency-domain watermarks and perceptual hashes are least robust to. All results are produced by `eval/natural_eval.py` and are fully reproducible (corpus via `eval/fetch_corpus.sh`).

Table 1: N1 (Kodak): watermark bit-error rate and exact-recovery rate vs. JPEG quality and rescale factor (24 images per condition).

JPEG q	scale	mean BER	exact-rec.
90	1.00	0.000	1.00
90	0.70	0.009	0.75
70	1.00	0.043	0.29
70	0.70	0.058	0.25
50	1.00	0.176	0.00
50	0.70	0.239	0.00
30	1.00	0.435	0.00
30	0.70	0.443	0.00

N1: watermark robustness. We embed a 64-bit identifier and measure bit-error rate (BER) and exact-recovery rate after JPEG recompression and rescaling (Table 1). On natural images the lightweight DWT-DCT-SVD watermark is reliable only at high quality (exact recovery 1.0 at $q=90$), degrades sharply by $q=70$ (exact recovery 0.29), and is effectively destroyed by $q=50$ (BER ≥ 0.16). It is *more* fragile on natural images than on smooth content, confirming that lightweight frequency-domain watermarking is unsuitable for social-media-grade compression and motivating learned schemes [2, 16] for the embedded path.

N2: durable recovery under recompression and scaling. Using a leave-one-out protocol, each registered image is transported (resize-to-70% + JPEG-40) and matched against the gallery; *mated* distances are to the correct entry and *impostor* distances to the nearest wrong entry. Matching content sits at mean Hamming distance 0.25/64 versus 22.67/64 for non-matching content—a wide separation yielding the ROC in Figure 1 (AUC = 1.0; equal-error rate at threshold 2, recovery 1.0 at false-match 0.0). Recovery thus succeeds on *natural* images after transport that destroys the watermark, establishing the fingerprint registry as the robust primary path for this transport class.

N3: crop sensitivity, and a crop-robust fix. We parameterise crop severity by the *retained center fraction* r (a crop with retained fraction r removes $1 - r$ of the area; thus $r = 0.70$ is a 30% crop), and measure nearest-neighbour recovery as a function of r for two methods (Table 2). The global perceptual hash is not crop-invariant and collapses accordingly: recovery is 0.75 at $r = 0.80$, 0.33 at $r = 0.70$, and essentially zero by $r = 0.50$. This confirms the central vulnerability—a system relying only on a global fingerprint is ineffective against an adversary who crops. We therefore add a local-feature path (ORB keypoints + RANSAC homography). It *resolves* the vulnerability over the practical range: recovery is 1.0 for all $r \geq 0.50$ (i.e. up to half the image removed), and degrades gracefully thereafter (0.92 at $r = 0.40$, 0.75 at $r = 0.30$) where the perceptual hash recovers nothing. Local features cost extra computation and per-image descriptor storage, and very aggressive crops ($r \leq 0.30$) and adversarial geometric warps remain open. We report the failure and its mitigation together.

7 Discussion

The asymmetry of provenance is its defining property: *signing is opt-in, verification is universal*. Only content a creator chose to sign carries credentials, but anyone may verify it without permission.

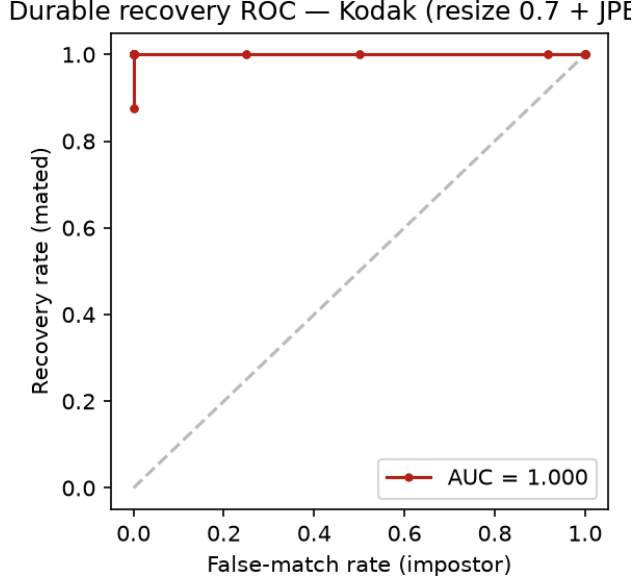


Figure 1: N2 (Kodak): ROC for fingerprint-based recovery (mated vs. impostor, leave-one-out) under resize-0.7 + JPEG-40 transport.

This makes “no credentials” the common case in the wild and is why honest messaging matters. Our results also delineate the open ecosystem’s boundary. Robust pixel watermarks deployed by major generative services (e.g. SynthID) are detectable *only* with proprietary, key-gated detectors; the open system cannot recover a stripped third-party asset’s provenance. Durable recovery here therefore protects content that passes through *participating* signers—a meaningful but bounded guarantee, and an argument for open, interoperable robust-watermarking standards.

8 Adoption and the Participation Model

A verifier is only as useful as the breadth of *signing*, so universal coverage is an adoption problem, not merely an engineering one. The system is designed to be adoptable by *any* entity—a person, an application, a camera, an AI generator, or a platform—and its design choices follow from that goal.

Interoperate, do not fork. We deliberately emit and consume *standard* C2PA rather than a private format. Content signed through our participant interface is readable by any C2PA verifier, and our verifier reads any C2PA asset regardless of signer, because it runs the reference toolkit. Adoption compounds only if implementations interoperate; a competing tag would fragment the very network effect it needs. This is the historical lesson of widely-adopted software substrates—value accrues to a common, re-implementable standard, not to private variants. As concrete evidence of the read direction, our verifier reports a *third-party* C2PA asset—signed by the reference `c2pa-rs` tooling, not by us—as VERIFIED with its original signer, tool, and declared edits; the emit direction is exercised by a sign-then-verify conformance round-trip.

Participation. Becoming a participant is a single signing operation at one’s point in the pipeline, beginning a chain of traceability there. The highest-leverage participants are platforms, whose

Table 2: N3 (Kodak): nearest-neighbour recovery rate vs. retained center fraction r (each crop followed by JPEG-70), for the global perceptual hash and the crop-robust ORB path.

retained r	pHash recovery	ORB recovery
1.00	1.00	1.00
0.90	1.00	1.00
0.80	0.75	1.00
0.70	0.33	1.00
0.60	0.12	1.00
0.50	0.04	1.00
0.40	0.00	0.92
0.30	0.08	0.75

dominant real-world behaviour today is to *strip* credentials on ingest; a conformant platform should instead *preserve* received credentials or re-sign on egress with the original chained as an ingredient.

A front door for any media. Coverage of “any image” is achieved by composing the layers into a single pipeline: C2PA verification, then—if no manifest is present—an attempt at durable recovery (§6), and otherwise an explicit UNKNOWN. The pipeline thus returns *a* result for every input while never fabricating a provenance verdict for unsigned content.

The irreducible boundary. No provenance system, however widely adopted, can verify a chain that was never created. Universal coverage is therefore approached by universal *signing* upstream, not by the verifier guessing downstream—which is exactly the unreliable detection problem provenance is meant to avoid. The honest terminal verdict UNKNOWN (not *fake*) is the price and the point of that discipline.

9 Limitations and Ethics

The evaluation uses the 24-image Kodak suite, a standard but modest scale that does not capture the collision pressure of a web-scale gallery: the per-comparison false-match probability compounds across many entries, so at scale the operating threshold must be tightened (or combined with the watermark) to hold the false-match rate. Our harness is dataset-agnostic—pointing it at a large corpus (e.g. DIV2K, ImageNet) is a one-line change—and larger, adversarial benchmarks remain future work. The global perceptual hash is not crop-invariant; the local-feature path mitigates this (Table 2), but very aggressive crops ($r \leq 0.30$) and adversarial geometric warps (rotation, perspective) are not yet robust, and local features add compute and per-image descriptor storage. The fingerprint path requires prior registration and is a fuzzy match. Offline revocation is unhandled: a revoked-but-still-chaining certificate currently reads as trusted. Ethically, the system is designed against two failure modes: over-trust (presenting an untrusted or merely-valid signature as endorsement) and the “liar’s dividend” (treating missing provenance as proof of fakery); both are explicitly precluded by the verdict model. Registries of fingerprints carry privacy implications that a deployment must address.

10 Reproducibility

All artifacts—verifier, signer, durable registry, and the evaluation harness that produced Tables 1–2 and Figure 1—are released under the Apache-2.0 license. The natural-image numbers are regenerated by `eval/fetch_corpus.sh` followed by `python3 eval/natural_eval.py`; library versions are pinned in the repository. Interoperability is checked by automated tests: the read direction asserts that a committed third-party-signed asset verifies (no network), and the browser fingerprint is asserted bit-for-bit against the reference implementation. The emit direction is captured by a sign-then-verify script on a network-connected host (signing requires a timestamp authority).

11 Future Work

The local-feature path closes the basic crop gap (Table 2); next is robustness to rotation and perspective warps and to very aggressive crops, e.g. via localized learned watermarks [16]. Beyond that: web-scale collision analysis and larger, adversarial natural-image and video benchmarks; integration of a learned watermark [2] for compression-robust embedded identifiers; offline revocation and trust-list distribution without a central authority; a decentralized, privacy-preserving recovery registry; and standardization of an open robust-watermark soft binding to close the third-party gap.

12 Conclusion

We have argued that the durable answer to synthetic media is not detection but verifiable, open, and *honestly-scoped* provenance, and built a system that embodies it: offline and gatekeeper-free verification, user-governed trust, a verdict model that refuses to conflate an intact chain with truth or to read absence as fakery, and a durable recovery layer that survives recompression, scaling, and—via local features—cropping, while staying explicit about where it does not. The same discipline shapes the adoption argument: interoperate with the standard the ecosystem already shares rather than fork it, lower the cost of becoming a signing participant, and let the front door return an honest UNKNOWN for the unsigned majority of media. Coverage of “almost any image” is then a function of how widely the world signs—an outcome this architecture is designed to enable, not to fake.

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